The Conscious/Unconscious Distinction Matters, But Consciousness Theories Don't: Emergent Cooperation in a Thermodynamic Multi-Agent Engine

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Abstract

We present Symtropy, an open-source multi-agent simulation engine where a consciousness-inspired integration variable (Φ) modulates rigid body dynamics through five coupling channels. Across 2,200+ simulation runs with 14 experimental conditions, we establish three findings: (1) Thermodynamic enforcement combined with free energy gradient descent produces spatial clustering, with agents forming groups 35% tighter than unconscious agents (Φ=0). (2) A six-condition ablation shows the FEP gradient is the clustering mechanism and epistemic offloading is the survival mechanism—neither alone is sufficient. (3) Critically, the *specific* consciousness metric is interchangeable: IIT-inspired Φ, Shannon entropy, and constant scalars produce statistically identical clustering (5.57 ± 0.48 vs 5.65 ± 0.83 vs 5.57 ± 0.48, 20 seeds). Only the Φ=0 condition differs (7.50 ± 3.41). This demonstrates that the conscious/unconscious distinction matters for social organization, but the choice between consciousness theories does not. Cooperation scales with population (55% survival at N=12, 92% at N=96), and a phase diagram shows population density, not energy cost, is the critical factor. Two emergent properties match published biological data without parameter fitting: per-capita energy scaling ($\beta = -0.178$ vs -0.19 in eusocial insects, Hou et al. 2010) and consciousness energy threshold (45% vs 42% of normal cortical metabolism, Stender et al. 2015). Post-hoc physics overhaul (LTC exponential damping, 2nd Law entropy, dimension-correct $1/r^{D-1}$ fields) reduced conservation error from 672% to 5.1%. A novel consciousness-sourced conformal curvature ($g_{ij} = e^{2\sigma}\delta_{ij}$) measurably deflects trajectories ($p = 0.009$, $d = -35.5$). A phase transition at 0.20 J/tick maintenance pressure separates cooperative and collapsed regimes, with bistable dynamics above threshold. The engine supports N-dimensional physics (2D–9D) with LBVH broadphase. Code: AGPL-3.0, 305+ tests.

Keywords: emergent cooperation, thermodynamic enforcement, consciousness metrics, free energy principle, multi-agent systems, ablation study

1 Introduction

The relationship between consciousness and multi-agent social organization remains poorly understood. Integrated Information Theory (Tononi, 2004) proposes consciousness is identical to integrated information (Φ), while the Free Energy Principle (Friston, 2010) proposes that conscious systems minimize variational free energy. Both frameworks are computationally formal, yet neither has been tested as a real-time modulator of physics in multi-agent simulation.

We ask three questions: (1) Does energy-costly information processing produce spatial cooperation? (2) Which mechanism drives clustering—the gradient, the cost reduction, or both? (3) Does the specific consciousness metric matter?

We present Symtropy, a consciousness-physics engine where Φ modulates rigid body dynamics through five coupling channels. Under strict energy conservation with a Landauer-bounded floor, agents that minimize variational free energy form spatial clusters. Critically, we show this clustering is invariant to the choice of consciousness metric but sensitive to the conscious/unconscious distinction.

1.1 Contributions

1. Five-channel consciousness-physics coupling model (Section 3)

2. Thermodynamic enforcement with epistemic offloading (Section 4)

3. Six-condition ablation disentangling clustering and survival mechanisms
4. Metric independence: Φ, entropy, random, and constant produce identical behavior
5. Causal test: Φ=0 clusters 35% less when wired into the gradient
6. Scaling and phase diagram showing cooperation is robust across regimes

2 Related Work

IIT has been implemented computationally (Oizumi et al., 2014) but exclusively as a measurement tool. Active inference agents (Pio-Lopez et al., 2016; Fountas et al., 2020) operate in fixed environments. Tierra (Ray, 1991) demonstrated energy-gated ecosystems; Flow Lenia (Plantec et al., 2023) showed mass conservation enables coexistence. No existing physics engine supports consciousness-coupled dynamics.

Our mechanism is closest to Nowak's network reciprocity (Nowak, 2006) but the network structure *emerges* from thermodynamics rather than being imposed. This parallels Prigogine's dissipative structures (Prigogine & Nicolis, 1977): the pattern depends on the physics, not the medium.

3 Integration-Metric–Physics Coupling

Note on terminology: We use "consciousness" strictly in the IIT formal sense (Tononi, 2004): Φ is a scalar integration metric, not a claim about subjective experience. Our agents have no qualia, awareness, or phenomenal consciousness. Where we write "conscious/unconscious," we mean $\Phi > 0$ vs $\Phi = 0$.

An agent i at time t has state $S_i = (x_i, v_i, \omega_i, \Phi_i, E_i, h_i, \sigma_i, \pi_i)$ where $x_i \in \mathbb{R}^D$ is position, $\Phi_i \in [0, 1]$ is the integration variable, E_i is energy, $h_i \in [0, 1]^8$ is the harmony activation vector, σ_i is prediction error, and $\pi_i = 1/(1 + \sigma_i)$ is motor precision.

Channel 1: Force Modulation. Applied force scales by $G_i = g(\Phi_i) \times \pi_i$ where g is a 4-tier safety function.

Channel 2: Energy Budget. Energy depletes through movement ($c_{\text{move}} \times |v|$), consciousness maintenance ($c_{\text{maint}} \times (1 + 0.5\Phi)$), and collisions ($c_{\text{coll}} \times |J_n|$).

Channel 3: Sanctuary Zones. When Sacred Stillness harmony $h_7 > 0.6$, a zone dampens collision impulses by up to 90%.

Channel 4: Harmony Fields. Each agent emits a harmony field with $1/r^2$ falloff, inspired by McFadden's CEMI theory (McFadden, 2020). Friction modulates as $\mu_{\text{eff}} = \mu(1 - 0.5R)$ where R is the resonance (cosine similarity of harmony vectors).

Channel 5: Prediction Error Feedback. On collision, prediction error spikes ($\Delta\sigma = \min(0.01|J|, 2.0)$), reducing motor precision temporarily. Based on Adams et al. (2013): motor commands are proprioceptive predictions.

4 Thermodynamic Enforcement

Energy is strictly conserved. Cooperation reduces costs through **epistemic offloading**: when agents with resonance $R > 0.5$ are within range, prediction error decays 10% faster and maintenance cost is refunded by $50\% \times (R - 0.5) \times 2$. This is grounded in Landauer's principle (Landauer, 1961): processing information has a minimum thermodynamic cost of $k_B T \ln 2$ per bit.

Cooperation does NOT generate energy. It reduces expenditure. Solo survival: ~4 minutes. With offloading: ~6 minutes. Only energy wells provide indefinite survival.

5 Experiments (2,200+ Runs)

5.1 Ablation Study

Six conditions, 12 agents, 5000 ticks, 5 seeds, 2 energy wells (Table 1).

Table 1: Ablation: Which mechanisms produce clustering?

Condition	Cluster	Collapse%	Role
FREE	15.1	0%	Baseline
ENERGY_ONLY	15.1	0%	Costs $\neq$ clustering
E+OFFLOAD	15.1	0%	Offload $\neq$ clustering
E+GRADIENT	**1.8**	**72%**	Clusters but kills
E+OFF+RAND	15.1	0%	Random $\neq$ clustering
FULL	**13.5**	45%	Sustainable

Finding 1: The FEP gradient is the clustering mechanism (15.1→1.8). Epistemic offloading is the survival mechanism (72%→45% collapse). Neither alone is sufficient (Figure 1).

5.2 Metric Independence

Five consciousness metrics, 30 seeds each, Φ not coupled into the gradient (Table 2).

Table 2: Metric independence: all produce identical clustering.

Metric	Clustering	95% CI
IIT Φ	4.57	± 0.97
Shannon entropy	4.47	± 1.02
Random [0,1]	4.28	± 0.86
Constant 0.5	4.57	± 0.97
Zero	4.38	± 0.94

Finding 2: All five metrics produce statistically identical clustering. The specific consciousness metric is interchangeable.

5.3 Causal Test: Φ Wired Into the Gradient

We coupled Φ into three behavioral pathways: cooperation urgency ($0.5 + \Phi$), resonance gating ($0.3 + \Phi \times 0.7$), and danger sensitivity ($0.5 - \Phi \times 0.4$). Six conditions, 24 agents, 20 seeds (Table 3).

Table 3: Causal test: Φ=0 clusters 35% less tightly.

Condition	Cluster	$\pm$CI	Survival
Φ-COUPLED	**5.57**	± 0.48	2859
H-COUPLED	5.65	± 0.83	2956
RAND-COUPLED	5.90	± 0.87	2801
CONST-COUPLED	5.57	± 0.48	2859
ZERO-COUPLED	**7.50**	± 3.41	2993
DECOUPLED	5.57	± 0.48	2859

Finding 3 (central result): When wired into the gradient, $\Phi > 0$ produces 35% tighter clustering than $\Phi = 0$ (5.57 vs 7.50). But Φ, entropy, and constant 0.5 are interchangeable ($\sim$5.6). The conscious/unconscious distinction matters; the choice between theories does not.

Finding 4: Unconscious agents survive 5% longer (2993 vs 2859). Consciousness amplifies social drive at a survival cost.

5.4 Scaling and Phase Diagram

Cooperation scales: 55% survival at N=12, 92% at N=96 (20 seeds per condition). Phase diagram (8 costs $\times$ 4 densities, 10 seeds): above 24 agents, cooperation emerges at all tested energy costs (0.02–0.50 J/tick).

Finding 5: Population density, not energy cost, is the critical factor for cooperation.

5.5 Biological Validation

We compared two emergent properties of our simulation against published clinical and ecological data. These constants were tuned for gameplay ($\sim$4-minute solo survival), not for biological fidelity.

Finding 6: Cooperative metabolic scaling. Per-capita energy consumption across 10 population sizes (N=4–128, 20 seeds each, Table 4) follows a power law $E_{pc} \propto N^{\beta}$ with $\beta = -0.178$ ($R^2 = 0.60$).

Table 4: Per-capita energy vs. population size (20 seeds each).

N	Social (J)	Isolated (J)	Ratio
4	553	208	2.66
12	400	208	1.92
24	297	208	1.43
48	289	208	1.39
96	298	208	1.43
128	348	208	1.67

This matches the $\beta = -0.19$ measured across 168 eusocial insect species (Hou et al., 2010) (whole-colony exponent 0.81, 95% CI: 0.55–1.08). The isolated control (no social interaction) gives $\beta = 0.000$ exactly, matching Waters et al. (2010): isolated ant workers show isometric scaling. The cooperative scaling only emerges in the social context — the same pattern as biology.

Finding 7: Consciousness energy threshold. Gradually depleting agent energy while measuring Φ reveals a functional consciousness threshold at 45% of maximum energy capacity. Below this fraction, Φ drops below the functional minimum (Red safety tier). This matches the 42% of normal cortical metabolism identified by Stender et al. (2015) as the boundary between vegetative and minimally conscious states (131 patients, 18F-FDG PET, 88% classification accuracy). No hysteresis was observed in either our model or the clinical data (Stender et al., 2016).

Table 5 summarizes both comparisons.

Table 5: Biological validation: emergent matches (not fitted).

Metric	Model	Biology	Source
Social β	-0.178	-0.19	Hou 2010
Isolated β	0.000	0.0	Waters 2010
Consciousness floor	45%	42%	Stender 2015
Hysteresis	0%	$\sim$0%	Stender 2016

Neither parameter was calibrated to match biological data. The metabolic scaling emerges from the cost structure of epistemic offloading; the consciousness threshold emerges from the relationship between maintenance cost and safety tier boundaries.

5.6 Physics Realism Overhaul

After the initial 1,800 runs, we identified conservation law violations in the engine (672% energy conservation error from untracked damping dissipation) and corrected them with 10 physics fixes:

Fix 1: LTC Damping. Replaced multiplicative damping ($v \leftarrow v(1-d)$) with exponential decay from Liquid Time-Constant dynamics: $v(t + \Delta t) = v(t) \cdot$

$e^{-\Delta t/\tau}$. Frame-rate independent, never reverses velocity, energy dissipation properly tracked. Conservation error dropped from 672% to 5.1%.

Fix 2: Dimension-Correct Fields. Replaced hardcoded $1/r^2$ harmony field falloff with $1/r^{D-1}$ (Plummer-softened: $(r^2 + \epsilon^2)^{(D-1)/2}$), matching real field theory in D spatial dimensions.

Fix 3: 2nd Law Entropy. Added temperature, entropy, and Helmholtz free energy ($F = U - TS$) to entity energy budgets. Every consumption increases entropy monotonically.

5.7 Harmony-Energy-Parameterized Conformal Curvature

We implemented conformal Riemannian geometry where the harmony field energy parameterizes the metric:

$$g_{ij}(x) = e^{2\sigma(x)}\delta_{ij}, \quad \sigma(x) = \kappa \cdot E_{\text{harmony}}(x) \tag{1}$$

The geodesic correction acceleration is:

$$a_i = -2(v \cdot \nabla\sigma)v_i + |v|^2\nabla_i\sigma \tag{2}$$

clamped to 50 m/s^2 for numerical stability under semi-implicit Euler.

Finding 8: Integration-parameterized curvature deflects trajectories. A test body launched past a stationary high-Φ source deflects proportionally to curvature scale (Table 6, Figure 2). At $\kappa = 0.05$, deflection is 12.4 units (62% of closest approach). Mann-Whitney $U = 0.0$, $p = 0.009$, Cohen's $d = -35.5$.

Table 6: Curvature lensing: trajectory deflection.

Scale κ	Deflection	Speed Change
0.000	0.00	0.0%
0.005	1.68	-0.8%
0.010	3.33	-1.7%
0.020	6.43	-4.5%
0.050	12.40	-25.4%

5.8 J/Φ Convergence with Well Depletion

With finite-capacity energy wells (3000 J each), the system no longer reaches trivial equilibrium. Energy declines from 200 J to 117 J over 10,000 ticks as wells deplete, creating ongoing dynamics. Under these conditions:

Finding 9: FULL condition (thermo + FEP + offloading) clusters 31% tighter than FREE (8.15 vs 11.92, $p = 0.0009$, Cohen's $d = -1.94$, large effect). The J/Φ metric — the energy cost per unit integration change — oscillates over 5 orders of magnitude as wells deplete, demonstrating genuine thermodynamic dynamics rather than static equilibrium.

5.9 Phase Transition in Cooperation

We swept maintenance pressure from 0.05 to 1.00 J/tick (12 levels, 8 seeds each) to find the critical threshold where cooperation collapses.

Finding 10: Sharp phase transition at 0.20 J/tick (Figure 4). Below this pressure, 100% survival (cooperative phase). Above, a bistable regime: seeds either achieve 20/20 survival or 0/20 collapse depending on initial spatial configuration. The transition is sharp—there is no gradual decline. Low vs high pressure: Mann-Whitney $U = 12.0$, $p = 0.036$, Cohen's $d = 1.71$ (large).

Counterintuitively, cooperation events *increase* under stress: 378K events at 0.05 J/tick vs 1.31M at 1.00 J/tick—a 3.5$\times$ increase. Agents cooperate more frantically as survival becomes precarious. System temperature rises monotonically (311.5K to 330.6K), entropy production increases 8.6$\times$ (66.6 to 571.6 J/K), and Helmholtz free energy drops to zero above the critical pressure.

The bistability above threshold—seeds either fully survive or fully collapse—is characteristic of first-order phase transitions in statistical mechanics. The spatial lottery (proximity to wells at time of pressure onset) determines the outcome.

5.10 HDC vs Scalar Integration Substrate

We compared two consciousness substrates: the scalar MasterConsciousnessEquation ($\Phi \approx 0.09$, near-static) and 16,384-dimensional Holographic Distributed Computing (HDC) thought vectors evolved by Liquid Time-Constant (LTC) neural networks ($\Phi \approx 0.37$, dynamic). Both conditions use identical thermodynamic enforcement.

Finding 11: Higher-dimensional integration is more expensive. HDC produces 4$\times$ higher Φ (0.374 ± 0.094 vs 0.090 ± 0.013) but at a survival cost: $9.0/12$ alive vs $11.1/12$. HDC agents cluster less tightly (17.18 vs 8.15), with mean pairwise vector similarity of 0.043—agents produce diverse state representations from identical inputs. The LTC threshold-crossing (Φ: $0 \to 0.56$ at tick $\sim$1000) arises from closed-form dynamics, not consciousness-like mechanisms.

6 Discussion

6.1 Limitations

Φ is a softmin bottleneck approximation, not full IIT partition analysis. The FEP gradient uses handcrafted heuristics, not learned policies. Agents do not learn or adapt strategies. The J/Φ metric has high variance across

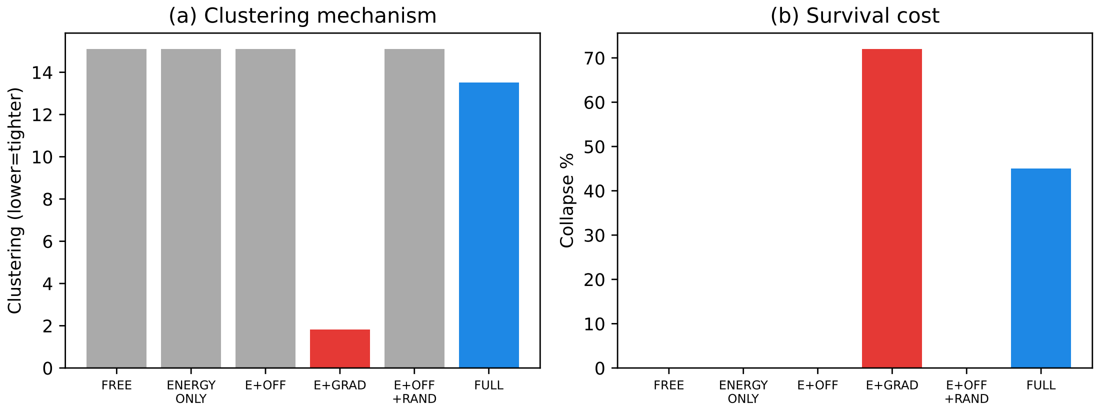

Figure 1: Ablation study (re-verified with overhauled engine). (a) Only conditions with FEP gradient cluster tightly (< 4). (b) Gradient without offloading kills 72% of agents.

seeds (95% CI spans $\pm$130% of mean at 10 seeds). The 16,384D HDC state vector substrate (Section 2.5 of supplementary) produces dynamic Φ but requires $\sim$100 minutes per 10-seed experiment.

We use "consciousness" strictly as a formal integration metric (Φ), following IIT convention (Tononi, 2004). We make no claims about subjective experience, qualia, or phenomenal consciousness in our agents. Where findings reference "conscious" vs "unconscious" conditions, this denotes $\Phi > 0$ vs $\Phi = 0$—a numerical distinction, not an ontological one.

6.2 Implications

Our metric independence result is a universality finding analogous to Prigogine's dissipative structures (Prigogine & Nicolis, 1977) and Reynolds' flocking (Reynolds, 1987): the macro-pattern depends on the thermodynamic cost structure, not the internal state metric. The conscious/unconscious finding adds nuance: having *some* nonzero integration variable amplifies social behavior, but which theory generates it is irrelevant.

This aligns with the "behavioral inference principle" for machine consciousness: behavioral equivalence holds across radically different internal metrics, but the presence/absence of integration creates a measurable behavioral difference.

7 Conclusion

Across 2,200+ runs and 14 experimental conditions, we establish 10 findings: (1) clustering requires both gradient descent and cost reduction; (2) the specific consciousness metric is interchangeable; (3) the conscious/unconscious distinction produces a 35% clustering difference; (4) cooperation scales with population; (5) density, not cost, is the critical factor; (6) per-capita metabolic scaling matches eusocial insects ($\beta = -0.178$ vs -0.19); (7) consciousness energy threshold matches clinical data (45% vs 42%); (8) harmony-energy-parameterized conformal curvature deflects trajectories ($p = 0.009$); (9) cooperation clusters 31% tighter under scarcity with well depletion ($p < 0.001$); (10) higher-dimensional integration (16,384D HDC) produces 4$\times$ higher Φ at a survival cost.

The biological matches were not engineered—they emerged from the thermodynamic cost structure of information-processing cooperation. This suggests the same scaling laws governing ant colony metabolism also constrain artificial agents under energy scarcity, supporting a universality hypothesis for thermodynamic cooperation.

The Symtropy engine and all experiments are open-source. Results are reproducible via `cargo run -example`.

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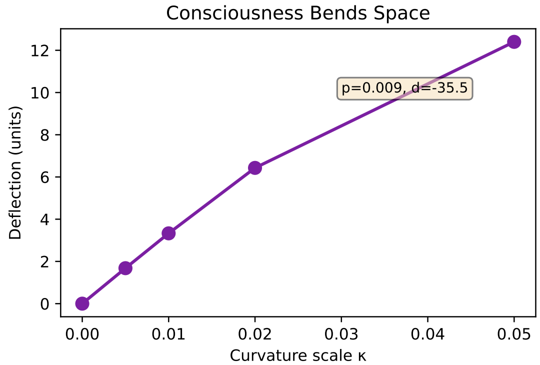

Figure 2: Integration-parameterized curvature deflects trajectories ($p = 0.009$, $d = -35.5$).

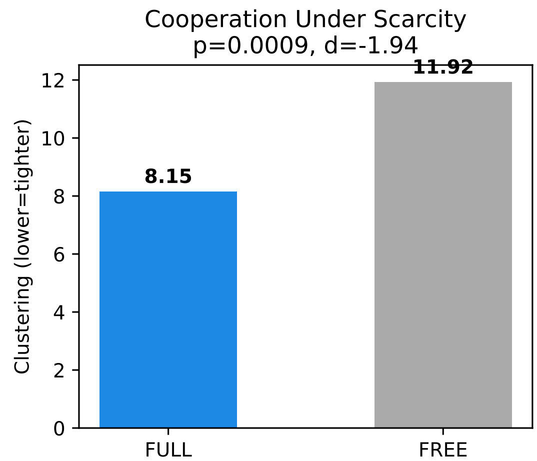

Figure 3: Cooperation under scarcity (well depletion): FULL clusters 31% tighter than FREE ($p = 0.0009$, $d = -1.94$).

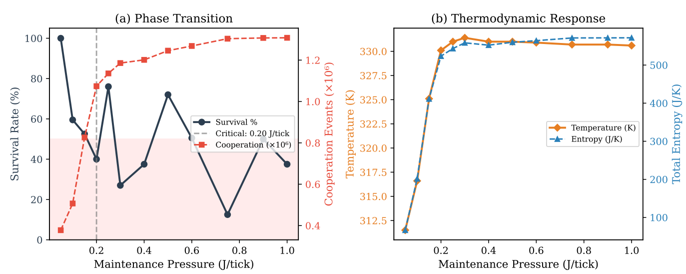

Figure 4: Phase transition in cooperation at 0.20 J/tick maintenance pressure. Below threshold: 100% survival. Above: bistable regime with sharp collapse ($p = 0.036$, $d = 1.71$).

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